01st Sep 2026

재생에너지 기술 및 응용

(Renewable Energy Technologies and Applications)

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Hydrogen **E**nerg**Y** Lab **ON E**arth

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Course Introduction

- **Lecture: Renewable Energy Technologies and Applications (L01211101)**
  - Classroom: 0413
  - Class time: Tuesday, 12:00–14:50

- **Professor**: Haewon Seo
  - Office: Prof Res Bldg , Rm 202
  - E–mail: heyone@hufs.ac.kr
  - Office hour: Monday (till 15:00)

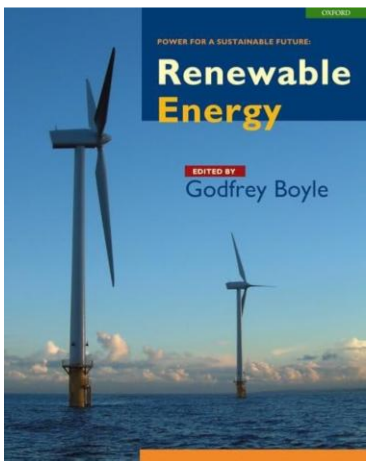
- **Scope**
  - This course explores the necessity of renewable energy in meeting the surging power demands of the AI era. Grounded in thermodynamics and fluid mechanics, it examines the operating principles and technologies of major renewable systems. Furthermore, it reviews the latest technological trends and discusses system integration strategies for practical applications.

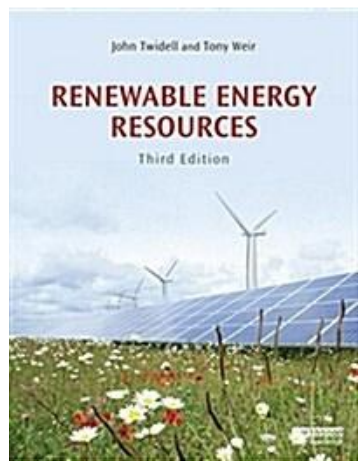
Course Introduction

❑ Textbook

– Main

- Renewable Energy (G. Boyle), 2^{nd} Ed., Oxford University Press (2004)
- Renewable Energy Sources, (J. Twidell & T. Weir), 3^{rd} Ed., Talyor & Francis (2015)
- Thermodynamics (Y.A. Cengel et al.), 9^{th} Ed., McGraw-Hill (2018)





– Reference

- Scientific articles
- IPCC, Corpernicus Climate Change, NOAA, NASA, etc.

Course Introduction

❑ **Schedule**; to be flexibly operated

Week	Date	Contents
01st	**Sep 01st 2026**	**Introduction to Energy (1)**
02nd	Sep 08th 2026	Introduction to Energy (2)
03rd	Sep 15th 2026	Fundamentals of Thermodynamics
04th	Sep 22nd 2026	Solar Thermal Energy
05th	Sep 29th 2026	Solar Photovoltaics (1)
06th	Oct 06th 2026	Solar Photovoltaics (2)
07th	Oct 13th 2026	Presentation (1)
08th	Oct 20th 2026	Midterm Exam
09th	Oct 27th 2026	Fundamentals of Fluid Dynamics
10th	Nov 03rd 2026	Wind Energy
11th	Nov 10th 2026	Hydropower
12th	Nov 17th 2026	Bioenergy
13th	Nov 24th 2026	Waste Energy & Geothermal Energy
14th	Dec 01st 2026	Presentation (2)
15th	Dec 15th 2026	Final Exam

Course Introduction

❑ Evaluation

- – Attendance (10%)
- – Assignment (20%)
- – Midterm Exam (30%)
- – Final Exam (40%)

1. Introduction to Energy (1)

Contents

What is Energy?

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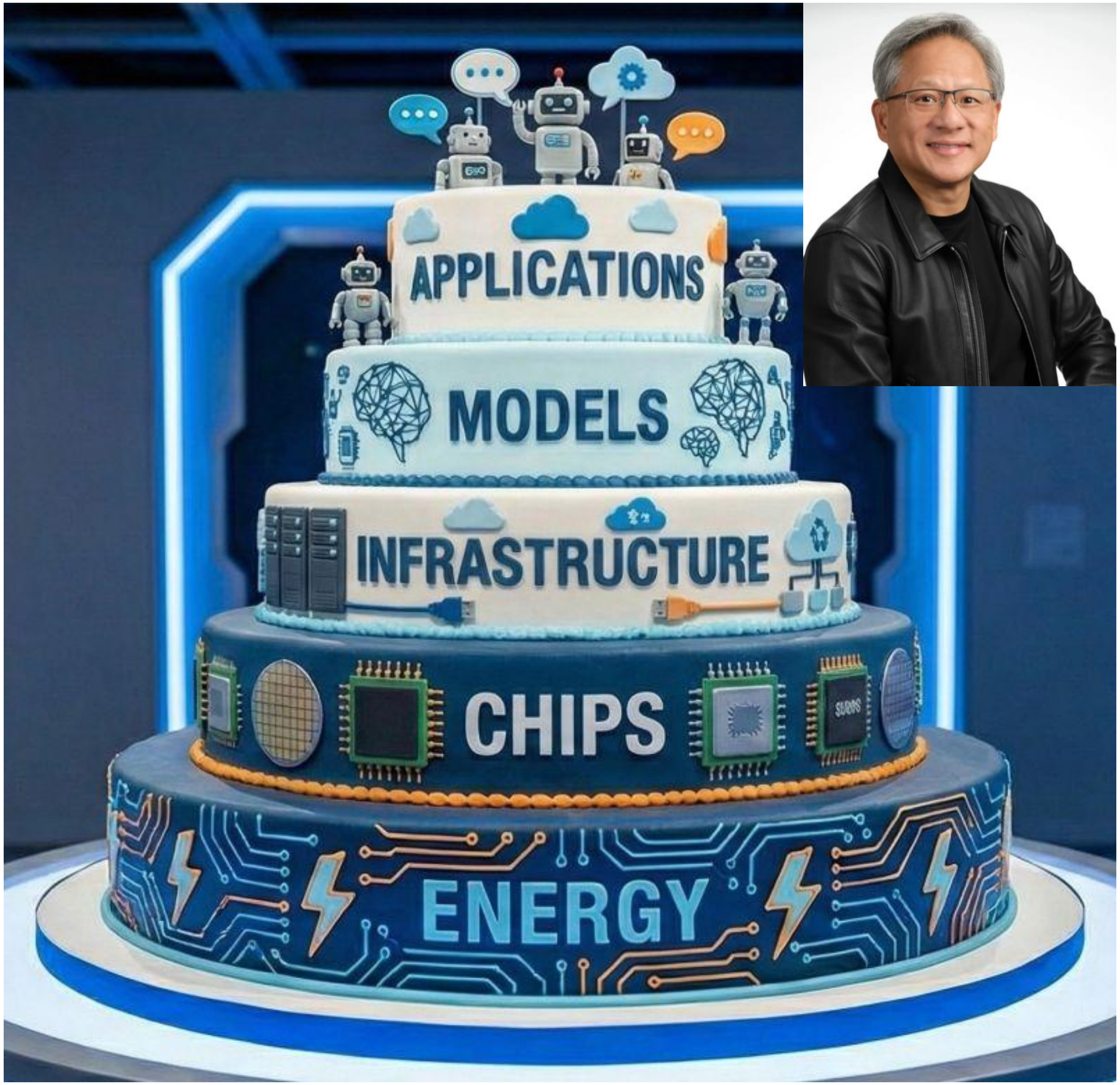
What is Energy?



What is Energy?

(*a*) Refrigerator

(*b*) Boats

(*c*) Aircraft and spacecraft

(*d*) Power plants

(*e*) Human body

(*f*) Cars

(*g*) Wind turbines

(*h*) Food processing

(*i*) A piping network in an industrial facility.

What is Energy?

❑ Energy

The quantitative property transferred to a system or body, recognizable as the capacity to do work, produce change, or emit heat and light

- **Energy is a scalar quantity (has magnitude, no direction)**

- **Total energy in a closed system is conserved over time**

$$\frac{dE_{\text{total}}}{dt} = 0 \Leftrightarrow E_{\text{total}} = \text{constant}$$

- **Units: J (SI), Wh, cal, eV, etc.**

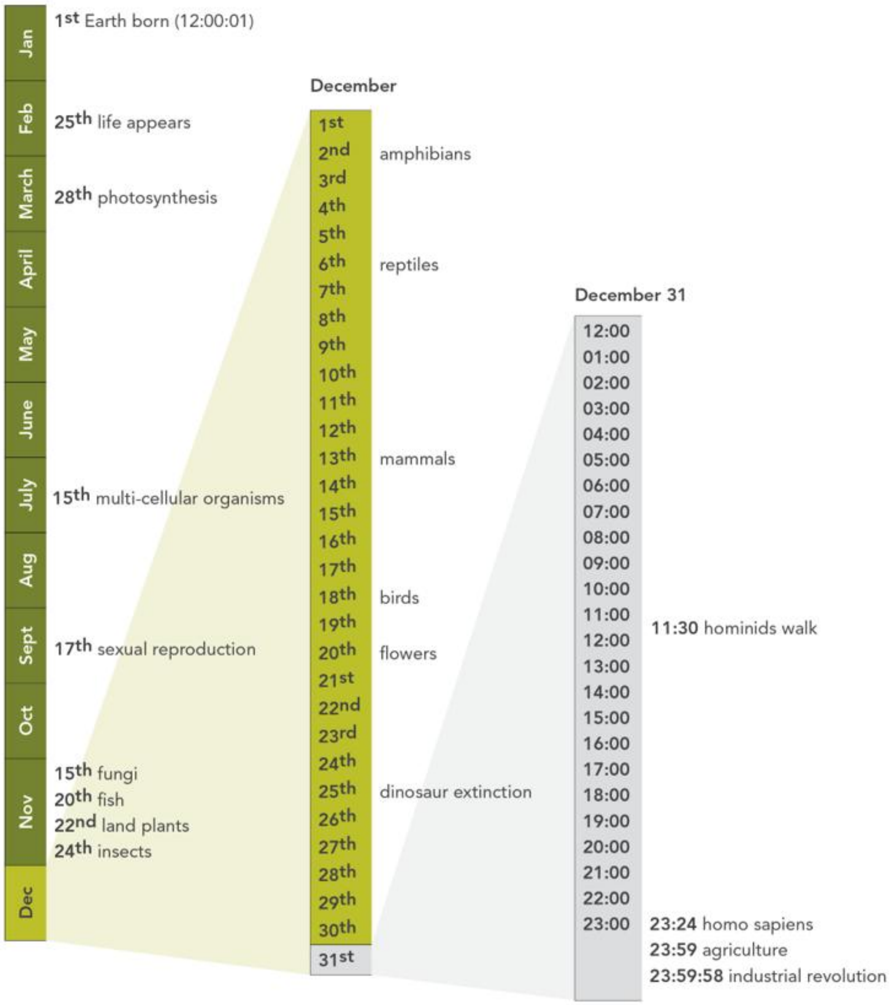
1 Wh = 3600 J (Electrical energy)

1 cal = 4.184 J (Metabolic energy)

1 eV = 1.602×10^{-19} J (Particle/Subatomic energy)

Timeline of Energy

❑ Geologic Calendar



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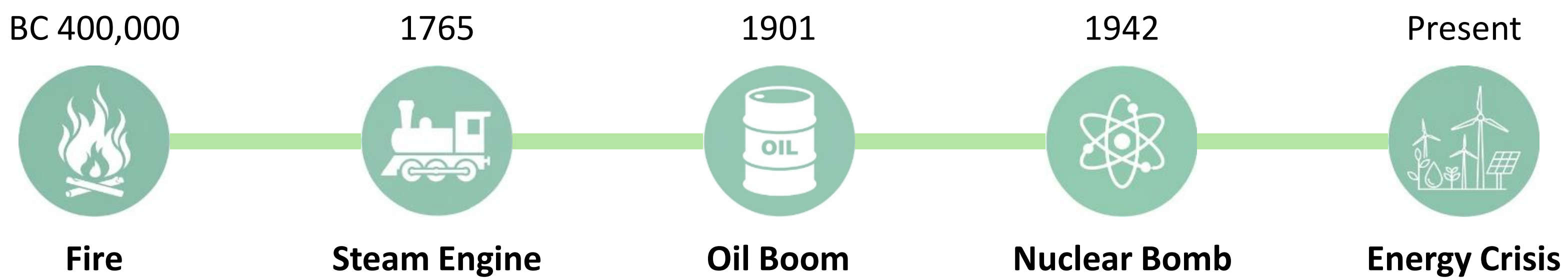
Timeline of Energy

❑ Energy Transition



Sir Benjamin Thompson (1753–1814)
The Nature of Heat (1798)
"Heat is a form of energy corresponding to a definite amount of mechanical work."

Julius Robert von Mayer (1814–1878)
The Laws of Conservation of Energy (1842)
"Energy can be neither created nor destroyed"

William John Macquorn Rankine (1820–1872)
Potential Energy (1853)

William Thomson, Lord Kelvin (1824–1907)
Kinetic Energy (1867)

Albert Einstein (1879–1955)
The Theory of Relativity (1905)
"Mass and energy are equivalent and interchangeable"

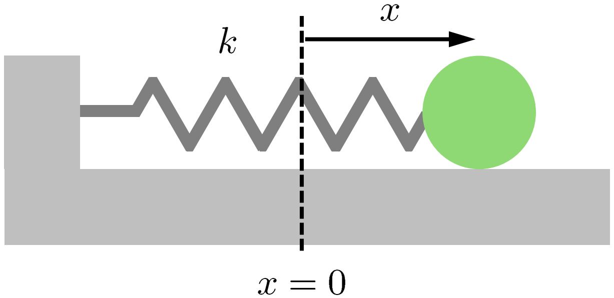
Forms of Energy

❑ Macroscopic Forms

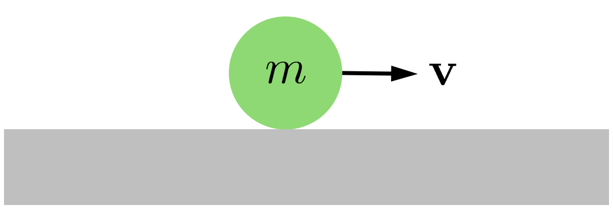
Possessed by a system as a whole relative to an external reference frame

(e.g.) Potential energy, Kinetic Energy

$$U = \frac{1}{2}kx^2$$



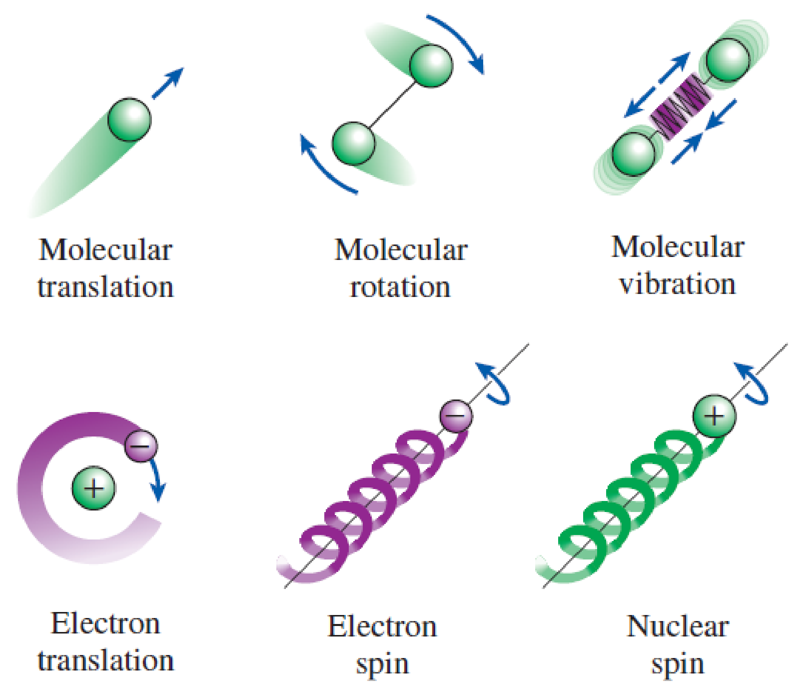
$$K = \frac{1}{2}mv^2$$



❑ Microscopic Forms

Related to the molecular structure and activity of a system, and independent of external reference frames.

The sum of all the microscopic forms of energy is called *the internal energy* of a system.



Forms of Energy

❑ Potential Energy

Stored energy that depends on the relative positions of objects

– Chemical Energy
(e.g., Gasoline, Food)

– Mechanical Energy
(e.g., Compressed spring)

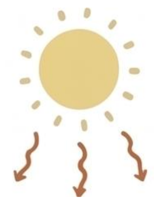
– Nuclear Energy
(e.g., Uranium fuel)

– Gravitational Energy
(e.g., Water behind a dam)

❑ Kinetic Energy

Energy that a system possesses as a result of its motion relative to some reference frame

– Radiant Energy
(e.g., Sunlight, Lamp light)



– Thermal Energy
(e.g., Boiling water)

– Motion Energy
(e.g., A moving car)

– Sound Energy
(e.g., A clapping sound)

International System of Units

SI units

A coherent system of units of measurement starting with seven base units, including **kg** (mass), **m** (length), **s** (time), **A** (electric current), **K** (temperature), **cd** (luminous intensity), and **mol** (amount of matter)

- **MLT dimension system**
  - – M ⇔ kilogram (kg), L ⇔ meter (m), T ⇔ second (s)
  - – (e.g.) [Velocity] = $L\ T^{-1}$ ⇔ $m\ s^{-1}$, [Volumetric flow] = $L^3\ T^{-1}$ ⇔ $m^3\ s^{-1}$

Variables	Unit (SI)	MLT
Acceleration	$m\ s^{-2}$	LT^{-2}
Angle	rad	$M^0L^0T^0$
Area	m^2	L^1
Density	$kg\ m^{-3}$	ML^{-3}
Energy	J	ML^2T^{-2}
Force	N	MLT^{-2}
Frequency	Hz	T^{-1}
Heat capacity	$J\ kg^{-1}\ K^{-1}$	$L^2T^{-2}\Theta^{-1}$
Length	m	L
Mass	kg	M

Variables	Unit (SI)	MLT
Power	W	ML^2T^{-3}
Pressure	Pa	$ML^{-1}T^{-2}$
Strain	–	$M^0L^0T^0$
Stress	Pa	$ML^{-1}T^{-2}$
Surface tension	$N\ m^{-1}$	MT^{-2}
Temperature	K	Θ
Time	s	T
Viscosity (dyn.)	Pa s	$ML^{-1}T^{-1}$
Viscosity (kin.)	$m^2\ s^{-1}$	L^2T^{-1}
Volume	m^3	L^3

International System of Units

❑ Metric Prefix

A unit prefix that precedes a basic unit of measure to indicate a multiple or submultiple of the unit

Multiple	Prefix	Adoption	Multiple	Prefix	Adoption
10^{30}	quetta, Q	2022	10^{-1}	deci, d	1795
10^{27}	ronna, R		10^{-2}	centi, c	
10^{24}	yotta, Y	1991	10^{-3}	mili, m	
10^{21}	zetta, Z		10^{-6}	micro, μ	1873
10^{18}	exa, E	1975	10^{-9}	nano, n	1960
10^{15}	peta, P		10^{-12}	pico, p	
10^{12}	tera, T	1960	10^{-15}	femto, f	1964
10^{9}	giga, G		10^{-18}	atto, a	
10^{6}	mega, M	1873	10^{-21}	zepto, z	1991
10^{3}	kilo, k	1795	10^{-24}	yocto, y	
10^{2}	hecto, h		10^{-27}	ronto, r	2022
10^{1}	deka, da		10^{-30}	quecto, q	

Force and Energy

❑ Force

Any action that tends to maintain or alter the motion of a body or to distort it

$$\mathbf{F} = \frac{d\mathbf{p}}{dt} = m\mathbf{a} \qquad \text{(Newton's 2}^{\text{nd}}\text{ Law)}$$

- **[Force] = [Mass][Acceleration] = $\mathrm{MLT^{-2}}$ ⇔ $\mathrm{kg\ m\ s^{-2}}$ = N**

  e.g., the gravitational forces on an 80.0-kg man on Earth and the Moon are respectively

  $F_{\mathrm{g,Earth}}$ = 80.0 kg × 9.81 m $\mathrm{s^{-1}}$ = 785 N = 80.0 kgf

  $F_{\mathrm{g,Moon}}$ = 80.0 kg × 1.62 m $\mathrm{s^{-1}}$ = 130 N = 13.1 kgf

❑ Energy

The capacity for performing work

$$W = \int \mathbf{F} \cdot d\mathbf{r}$$

- **[Energy] = [Force][Length] = $\mathrm{ML^2T^{-2}}$ ⇔ $\mathrm{kg\ m^2\ s^{-2}}$ = J**

  e.g., the kinetic energy of an 80.0-kg man who runs at 10.0 m $\mathrm{s^{-1}}$ is

  K = 1/2 × 80.0 kg × (10.0 m $\mathrm{s^{-1}})^2$ = 4000 J (c.f., daily adult caloric intake = ~10 MJ)

Power and Pressure

❑ Power

The rate at which work is performed

$$P = \frac{dW}{dt}$$

- [Power] = [Energy][Time]$^{-1}$ = ML^2T^{-3} ⇔ J s^{-1} = W

  e.g., the power an 80.0-kg man exerts climbing stairs at 1.00 m s^{-1} is

  P = 80.0 kg × 9.81 m s^{-2} × 1.00 m s^{-1} = 785 W   (c.f., 1 horsepower = 746 W)

❑ Pressure

The perpendicular force per unit area, or the stress at a point within a confined fluid

$$p = \frac{F}{A}$$

- [Pressure] = [Force][Area]$^{-2}$ = $ML^{-1}T^{-2}$ ⇔ N m^{-2} = Pa

  e.g., the pressure exerted on the ground by an 80.0-kg man with a total contact area of 0.0450 m^2 is

  p = 80.0 kg × 9.81 m s^{-2} / 0.0450 m^2 = 17.4 kPa   (c.f., p_{atm} = 101.325 kPa)

Power Needs of a Car to Climb a Hill

❑ Quiz (1)

A man pushes a 100-kg cart, including its contents, up a ramp that is inclined at an angle of 30° from the horizontal. The local gravitational acceleration is 9.8 m s^{-2}. Considering the cart and its contents as the system, determine the work, in kJ, needed to move it along this ramp a distance of 100 m.

$$
\begin{aligned}
W &= mgl\sin\vartheta \\
&= 100\ \text{kg} \times 9.8\ \text{m s}^{-2} \times \sin(30°) \\
&= \qquad \textbf{kJ}
\end{aligned}
$$

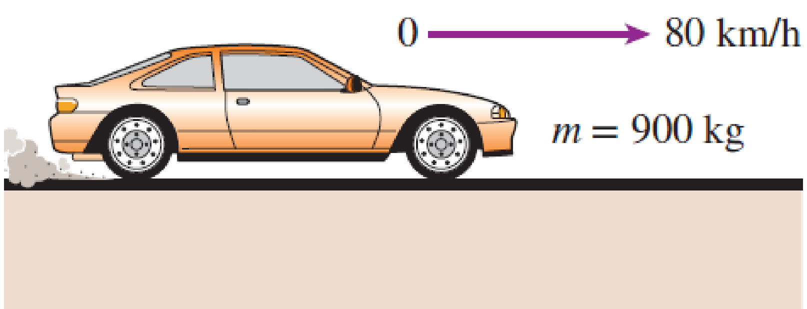
Power Needs of a Car to Accelerate

❑ Quiz (2)

Determine the power required to accelerate a 900-kg car shown in from rest to a velocity of 80 km h^{-1} in 20 s on a level road

$K = (1/2)m(v_f^2 - v_i^2)$
$= (1/2) \times 900\ \text{kg} \times (\qquad - 0^2)\ \text{m}^2\ \text{s}^{-2}$
$=$ **kJ**

$P = K/\Delta t$
$=$ **W**



Power Generation

❑ Quiz (3)

A household is paying KRW 215 kWh^{-1} for electric power. To reduce its power bill, the household installed a solar panel with a rated power of 3 kW. If the solar panel operates 1400 hours per year (~4 hours per day) at the rated power, determine *the amount of electric power generated by the solar panel* and *the money saved by the household per year*.

Total energy = Energy per unit time × Time interval
=
= **$kWh\ yr^{-1}$**

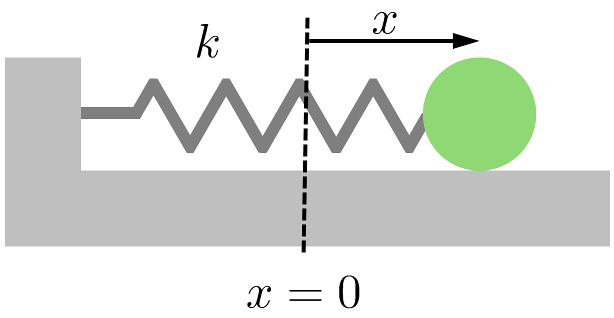
Money saved = Total energy × Unit cost of energy
=
= **KRW yr^{-1}**

Assignment (~8th Sep)

❑ Energy Conservation

Consider a one-dimensional motion along the *x*-axis of a spring–mass system, where a ball of mass *m* is attached to a spring with a spring constant *k*. Assume that no nonconservative forces are acting on the system. Solve the following problems:

*friction, air resistance



- Write down *the equation of motion* for the system.

- Determine *the angular frequency ω of the oscillation* using *k* and *m*.

Assignment (~8th Sep)

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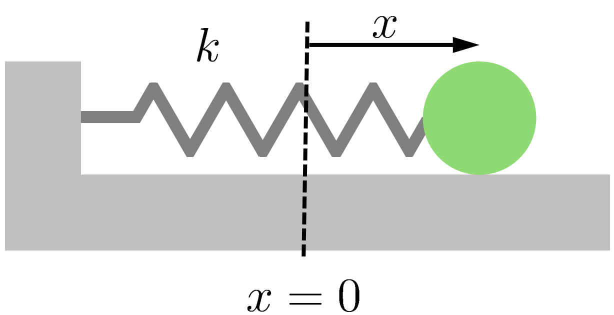

- Derive *the displacement x(t)* using the amplitude x_0, *k*, and *m.*

- Estimate *the total energy of the system* E_{total} and prove that it is constant over time.

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